Algebraic Topology

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0 Abstract Algebra Prerequisites

0.1 Group Theory Basics

Definition 0.1.1. Group

A **group** $\langle G, * \rangle$ is a set G, closed under a binary operaton $*$, which satisfies

- associativity of $*$
- existence of identity element e
- existence of inverse element a' of a for all $a \in G$

Definition 0.1.2. Permutation and Symmetric Group

Let $A \neq \varnothing$. A **permutation** of A is a bijection $\sigma \colon A \to A$. The **symmetric group** on A is

$$S_A = \mathrm{Sym}(A) = \{\sigma \colon A \to A \mid \sigma \text{ is bijective}\}$$

For $A = \{1, 2, \ldots, n\}$, denote $S_n = S_{\{1,\ldots,n\}}$ with order $|S_n| = n!$.

Definition 0.1.3. Cycle and Transposition

A **k-cycle** $(a_1\ a_2\ \cdots\ a_k) \in S_n$ $(k \geq 2)$ is defined by

$$\sigma(a_i) = a_{i+1}\ (1 \leq i < k), \quad \sigma(a_k) = a_1, \quad \sigma(x) = x\ \forall x \notin \{a_1, \ldots, a_k\}.$$

A 2-cycle is a **transposition**. Inverses satisfy:

$$(a_1\ a_2\ \cdots\ a_k)^{-1} = (a_k\ a_{k-1}\ \cdots\ a_1).$$

Proposition 0.1.4. Let $\mathrm{supp}(\sigma) = \{x \in \{1, \ldots, n\} \mid \sigma(x) \neq x\}$.

(a) **Disjoint Commutativity**: $\mathrm{supp}(\sigma) \cap \mathrm{supp}(\tau) = \varnothing \implies \sigma\tau = \tau\sigma$.

(b) **Disjoint Cycle Decomposition**: $\forall \pi \in S_n$, $\exists!$ partition of moved elements into disjoint cycles $c_1, \ldots, c_m$ such that

$$\pi = c_1 c_2 \cdots c_m.$$

Definition 0.1.5. General Linear Group

For $n \in \mathbb{Z}^+$,

$$\mathrm{GL}(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\} = M_n(\mathbb{R})^\times.$$

$(\mathrm{GL}(n, \mathbb{R}), \cdot)$ is nonabelian for all $n \geq 2$.

Definition 0.1.6. Dihedral Group

For $n \geq 3$, the **dihedral group** D_n is the isometry group of a regular n-gon in $\mathbb{R}^2$:

$$D_n = \langle \rho, \mu \mid \rho^n = e,\ \mu^2 = e,\ \mu\rho\mu = \rho^{-1} \rangle.$$

Theorem 0.1.7. Let $n \geq 3$. Then $|D_n| = 2n$, with canonical form:

$$D_n = \left\{\rho^k, \mu\rho^k \mid 0 \leq k < n\right\},$$

governed by:

$$\rho^k\mu = \mu\rho^{-k} = \mu\rho^{n-k}, \quad (\rho^k)^{-1} = \rho^{n-k}, \quad (\mu\rho^k)^{-1} = \mu\rho^k.$$

Definition 0.1.8. Subgroup

Let $(G, \cdot)$ be a group. $H \subseteq G$ is a **subgroup** ($H \leq G$) if

$$H \neq \varnothing \quad \text{and} \quad (H, \cdot|_{H\times H}) \text{ is a group.}$$

$H < G$ denotes a proper subgroup ($H \neq G$). $H = \{e\}$ is the **trivial subgroup**; $H = G$ is the **improper subgroup**.

Theorem 0.1.9 (Subgroup Criteria)**.** Let G be a group and $H \subseteq G$.

(a) **Three-Step Test**:

$$H \leq G \iff (\forall a, b \in H \implies ab \in H) \wedge (e \in H) \wedge (\forall a \in H \implies a^{-1} \in H).$$

(b) **One-Step Test**:

$$H \leq G \iff H \neq \varnothing \wedge (\forall a, b \in H \implies ab^{-1} \in H).$$

(c) **Finite Subgroup Test**: If $|H| < \infty$ and $H \neq \varnothing$,

$$H \leq G \iff (\forall a, b \in H \implies ab \in H).$$

Example 0.1.10 (Standard Subgroups)**.**

(a) **Center**: $Z(G) = \{z \in G \mid zg = gz \ \forall g \in G\} \leq G$ is abelian.
(b) **Special Linear Group**: $\mathrm{SL}(n, \mathbb{R}) = \{A \mid \det(A) = 1\} \leq \mathrm{GL}(n, \mathbb{R})$.
(c) **Roots of Unity**: $U_n = \{z \in \mathbb{C} \mid z^n = 1\} \leq U = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$.

Definition 0.1.11. Cyclic Subgroup and Generator

For $a \in G$, the **cyclic subgroup generated by** a is

$$\langle a\rangle = \{a^n \mid n \in \mathbb{Z}\} \leq G \quad (\text{in additive notation, } \{na \mid n \in \mathbb{Z}\}).$$

G is **cyclic** if $\exists a \in G$ such that $G = \langle a\rangle$; a is called a **generator** of G.

Theorem 0.1.12. $G = \langle a\rangle \implies G$ is abelian.

Theorem 0.1.13 (Division Algorithm for $\mathbb{Z}$)**.**

$$\forall m \in \mathbb{Z}^+, \forall n \in \mathbb{Z}, \quad \exists!\, q, r \in \mathbb{Z} \text{ such that } n = mq + r \quad \text{and} \quad 0 \leq r < m.$$

Theorem 0.1.14 (Subgroups of Cyclic Groups)**.** Let $G = \langle a \rangle$.

(a) $H \leq G \implies H$ is cyclic.
(b) The subgroups of $(\mathbb{Z}, +)$ are precisely $n\mathbb{Z} = \langle n \rangle$ for $n \in \mathbb{Z}_{\geq 0}$.

Definition 0.1.15. Greatest Common Divisor

For $r \in \mathbb{Z}^+, s \in \mathbb{Z}_{\geq 0}$, $\gcd(r, s)$ is the unique positive generator of $r\mathbb{Z} + s\mathbb{Z}$:

$$\langle \gcd(r,s) \rangle = \{nr + ms \mid n, m \in \mathbb{Z}\} \implies \exists n, m \in \mathbb{Z} \colon \gcd(r,s) = nr + ms.$$

Theorem 0.1.16 (Classification of Cyclic Groups)**.** Let $G = \langle a \rangle$.

(a) $|G| = \infty \implies G \cong (\mathbb{Z}, +)$ via $a^k \mapsto k$.
(b) $|G| = n < \infty \implies G \cong (\mathbb{Z}_n, +_n)$ via $a^k \mapsto k \pmod{n}$.

Theorem 0.1.17 (Orders, Generators, and Subgroup Lattice)**.** Let $G = \langle a \rangle$ with $|G| = n$, and let $b = a^s \in G$.

(a) **Element Order**:

$$|a^s| = |\langle a^s \rangle| = \frac{n}{\gcd(n,s)}, \quad \text{and} \quad \langle a^s \rangle = \langle a^{\gcd(n,s)} \rangle.$$

(b) **Generators**:

$$\langle a^s \rangle = G \iff \gcd(n,s) = 1.$$

The number of distinct generators of G is $\phi(n) = |\{k \in \{1, \ldots, n\} \mid \gcd(n,k) = 1\}|$.
(c) **Lagrange Divisibility**: $H \leq G \implies |H| \mid n$.
(d) **Subgroup Uniqueness**: For each divisor $d \mid n$ $(d > 0)$, there exists a **unique** subgroup of order d:

$$H_d = \langle a^{n/d} \rangle = \left\{ x \in G \mid x^d = e \right\}, \quad |H_d| = d.$$

Theorem 0.1.18 (Subgroup Intersections)**.** Let $\{H_i\}_{i \in I}$ be any nonempty collection of subgroups of G. Then

$$\bigcap_{i \in I} H_i \leq G.$$

Definition 0.1.19. Subgroup Generated by a Subset

For $\varnothing \neq S \subseteq G$:

$$\langle S \rangle = \bigcap \{H \leq G \mid S \subseteq H\}.$$

S is a **generating set** of G if $\langle S \rangle = G$. If $\exists S \subseteq G$ with $|S| < \infty$ and $\langle S \rangle = G$, G is **finitely generated**.

Theorem 0.1.20 (Constructive Characterization)**.** For $\varnothing \neq S \subseteq G$:

$$\langle S \rangle = \{x_1^{\epsilon_1} x_2^{\epsilon_2} \cdots x_k^{\epsilon_k} \mid k \in \mathbb{N},\, x_j \in S,\, \epsilon_j \in \{\pm 1\}\}.$$

Definition 0.1.21. Cayley Digraph

Let G be a finite group and $S \subseteq G$ a generating set. The **Cayley digraph** $\Gamma(G, S) = (V, E)$ is defined by:

- $V = G$;
- $E = \{(g, gs; s) \mid g \in G,\ s \in S\}$, representing a directed arc $g \xrightarrow{s} gs$.

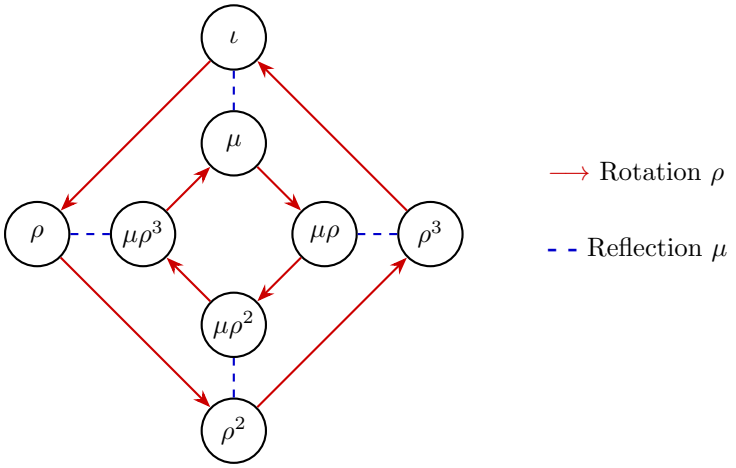
When $s = s^{-1}$, (g, gs) and (gs, g) collapse to an undirected edge.

Proposition 0.1.22 (Structural Properties of $\Gamma(G, S)$)**.**

(a) **Connectivity**: $\Gamma(G, S)$ is connected $\iff \langle S \rangle = G$.
(b) **Valency**: $\forall g \in G,\ \forall s \in S,\ \deg^{(s)}_{\text{in}}(g) = \deg^{(s)}_{\text{out}}(g) = 1$.
(c) **Vertex Transitivity**: $G \leq \text{Aut}(\Gamma(G, S))$ via left regular action $\lambda_h(g) = hg$.
(d) **Cycles and Relations**: Directed closed paths from e:

$$e \xrightarrow{s_1^{\epsilon_1}} \cdots \xrightarrow{s_k^{\epsilon_k}} e \iff s_1^{\epsilon_1} \cdots s_k^{\epsilon_k} = e \text{ in } G.$$

Example 0.1.23. Below is a Cayley digraph of D_4 with generating set $S = \{\rho, \mu\}$



Exercise 0.1.24. Let G be an arbitrary group and let $a, b \in G$.

- Prove that if $|ab| = n < \infty$, then $|ba| = n$.

  Proof. Notice

  $$(ab)^n = a(ba)^{n-1}b = e \implies a(ba)^n = ea = a \implies (ba)^n = e \implies |ba| \leq n$$

  and

  $$((ba)^m = e \implies (ab)^m = e) \implies |ba| \geq |ab| = n$$

  Then $|ba| = n$. □

- Deduce that for any $g \in G$ and $x \in G$, $|x| = |gxg^{-1}|$

Proof. We have

$$(gxg^{-1})^m = gx^m g^{-1} = e \implies gx^m = g \implies x^m = e \implies \left|gxg^{-1}\right| \geq |x|$$

We also have

$$x^m = e \implies gx^m g^{-1} = e \implies gxg^{-1})^m = e \implies |x| \geq \left|gxg^{-1}\right|$$

Therefore, $|x| = |gxg^{-1}|$. □

Remark. In fact, let $a = gx$ and $b = g^{-1}$ then $|ab| = |gxg^{-1}| = |ba| = |x|$.

Exercise 0.1.25. Let H and K be subgroups of a group G, and define the subset $HK = \{hk : h \in H, k \in K\}$

- Prove that if G is abelian, then $HK \leq G$.

  Proof. Let $hk, h'k' \in HK$, then we have

  $$(hk)(h'k')^{-1} = hkk'^{-1}h'^{-1} = (hh'^{-1})(kk'^{-1}) \in HK$$

  implying that $HK \leq G$. □

- Give a counterexample when G is nonabelian.

  Proof. Let $G = D_3, H = \langle\mu\rangle, K = \langle\mu\rho\rangle$, then $HK = \{e, \mu, \rho, \mu\rho\}$ is not a subgroup of G. □

Exercise 0.1.26. Let G be a finite cyclic group. Prove that if $H, K \leq G$ with $H \neq K$, then $|H| \neq |K|$.

Proof. Let $G = \langle a\rangle$ and $|G| = n$. Then $H, K \leq G$ implies $H = \langle a^p\rangle = \langle a^{\gcd(n,p)}\rangle, K = \langle a^q\rangle = \langle a^{\gcd(n,q)}\rangle$ for some $p, q \in \mathbb{Z}^+$, then

$$H \neq K \implies \gcd(n,p) \neq \gcd(n,q) \implies |H| = \frac{n}{\gcd(n,p)} \neq \frac{n}{\gcd(n,q)} = |K|$$

□

Exercise 0.1.27. Prove that a group G with $|G| > 1$ has no proper nontrivial subgroups if and only if G is finite and is isomorphic to $\mathbb{Z}_p$ for some prime p.

Proof. $\implies$: Let $a \neq e \in G$.. Then $\langle a\rangle \leq G$ must equals G for otherwise it will be a proper nontrivial subgroup of G. Then G is cyclic. If G is infinite, then $\langle a\rangle = \langle a^2\rangle = G$ leads to a contradiction since $a \notin \langle a^2\rangle$. If $|G| = mn$ for some $m, n > 1$, then $|a^m| = mn/\gcd(mn, m) = n$ implying that $\langle a^m\rangle$ is a proper nontrivial subgroup of G. Therefore, $G = \langle a\rangle$ and $|G| = p$ with p prime. It follows that $G \cong \mathbb{Z}_p$.

$\impliedby$: Let $G \cong \mathbb{Z}_p$, then $G = \langle a\rangle$ and $|G| = p$. If $H \leq G$, then $H = \langle a^s\rangle = \langle a^{\gcd(p,s)}\rangle$. Since p is prime, we know $H = \{e\}$ or $H = G$. □

1 Glossary

Abstract Algebra Prerequisites

Group

Permutation and Symmetric Group

Cycle and Transposition

General Linear Group

Dihedral Group

Subgroup

Cyclic Subgroup and Generator

Greatest Common Divisor

Subgroup Generated by a Subset

Cayley Digraph